Problem Set #34

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Problem 8

(e)

$$y = \arccos\left(\frac{x-3}{2x-1}\right)$$

$$\begin{aligned} \frac{x-3}{2x-1} &\geq -1 \\ \frac{x-3}{2x-1} + \frac{2x-1}{2x-1} &\geq 0 \\ \frac{3x-4}{2x-1} &\geq 0 \end{aligned}$$

$$\begin{aligned} 2x-1 &= 0 \\ x &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} 3x-4 &= 0 \\ x &= \frac{4}{3} \end{aligned}$$

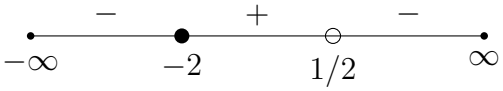


$$x \in \left(-\infty, \frac{1}{2}\right) \cup \left[\frac{4}{3}, \infty\right)$$

$$\begin{aligned} \frac{x-3}{2x-1} &\leq 1 \\ \frac{x-3}{2x-1} - \frac{2x-1}{2x-1} &\leq 0 \\ \frac{-x-2}{2x-1} &\leq 0 \end{aligned}$$

$$\begin{aligned} 2x-1 &= 0 \\ x &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} -x-2 &= 0 \\ x &= -2 \end{aligned}$$



$$x \in (-\infty, -2] \cup \left(\frac{1}{2}, \infty\right)$$

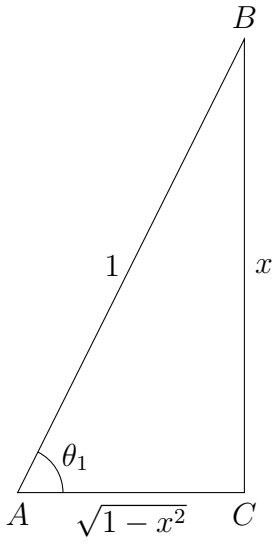
$$x \in \left(\left(-\infty, \frac{1}{2}\right) \cup \left[\frac{4}{3}, \infty\right)\right) \cap \left((-\infty, -2] \cup \left(\frac{1}{2}, \infty\right)\right)$$

$$\boxed{x \in (-\infty, -2] \cup \left[\frac{4}{3}, \infty\right)}$$

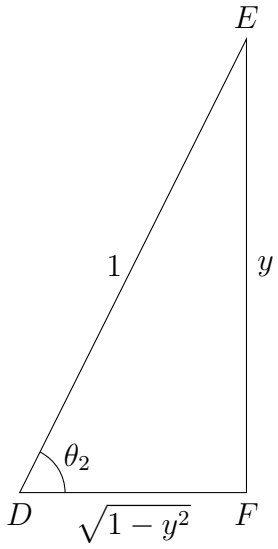
Problem 9

Theorem. $\arcsin x - \arcsin y = \arcsin\left(x\sqrt{1-y^2} - y\sqrt{1-x^2}\right)$

Proof.



$$\begin{aligned}
\text{Let } \theta_1 &= \arcsin x \\
\sin\theta_1 &= \sin(\arcsin x) \\
\frac{\text{opp}}{\text{hyp}} &= x \\
\frac{BC}{AB} &= x \\
\text{Let } BC = x,\ AB &= 1 \\
AC &= \sqrt{1-x^2} \\
\\
\sin\theta_1 &= x \\
\cos\theta_1 &= \sqrt{1-x^2}
\end{aligned}$$



$$\begin{aligned}
\text{Let } \theta_2 &= \arcsin y \\
\sin\theta_2 &= \sin(\arcsin y) \\
\frac{\text{opp}}{\text{hyp}} &= y \\
\frac{EF}{DE} &= y \\
\text{Let } EF = y,\ DE &= 1 \\
DF &= \sqrt{1-y^2} \\
\\
\sin\theta_2 &= y \\
\cos\theta_2 &= \sqrt{1-y^2}
\end{aligned}$$

$$\begin{aligned}
\text{RHS: } &\arcsin\left(x\sqrt{1-y^2} - y\sqrt{1-x^2}\right) \\
&= \arcsin\left(\sin\theta_1\cos\theta_2 - \sin\theta_2\cos\theta_1\right) \\
&= \arcsin\left(\sin\left(\theta_1 - \theta_2\right)\right) \\
&= \theta_1 - \theta_2 \\
&= \arcsin x - \arcsin y
\end{aligned}$$

□